

How Electricity Markets Work

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Part I

Introduction to Wholesale Electricity Markets

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The Electrical Grid

- Systems we rely on every day:
 - The internet
 - The water supply network
 - The electrical grid
- When these systems work, we often take them for granted
- When they fail, we realize how dramatically they improve our quality of life
- What happens when you turn a light on?
 - Over a hundred years of human innovation kick in
 - Transmission and distribution lines supply more current
 - Generators work harder to supply that extra power

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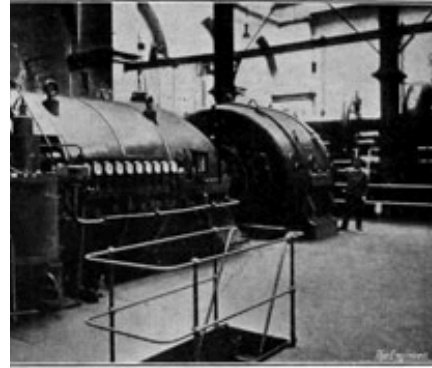
The Electrical Grid – A Brief History

- Late 1800s – early 1900s in US and UK
- Supply and demand initially co-located
- Private utility companies formed
- Some utilities began interconnecting their systems
- Nationalization in UK and France: regional grids → nationwide grids
- By 1960: nationwide grids across much of Europe

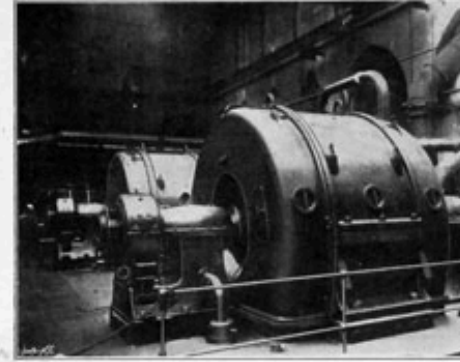
Photo from Grace's Guide To British Industrial History.

THE DEPTFORD ELECTRICITY GENERATING STATION

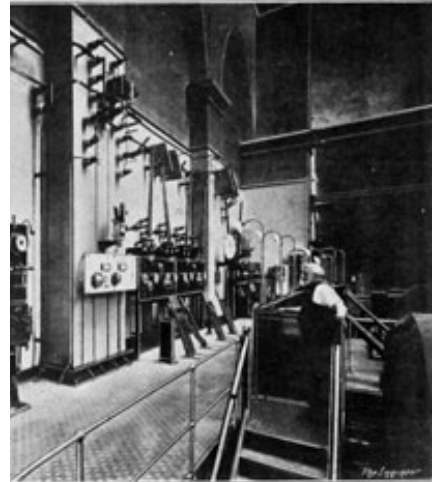
(For description see page 110)



HORSE-POWER TURBO-ALTERNATOR



10,000 HORSE-POWER TURBO-ALTERNATOR



SWITCHES AND MAIN-CONTROLLING SWITCHES



THE NEW HIGH-TENSION SWITCHBO

The Wholesale Electricity Market

AKA Power Exchange (PX)

- Balances supply and demand at all times
- Minimizes cost of electricity production and maximizes value to consumers
- Market participants:
 - Loads
 - **Bid** to buy power
 - Utility companies and large industrial consumers
 - Generators
 - **Offer** to sell power
 - Power station (e.g., hydro), wind/solar farm, etc.

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Wholesale Electricity Market Operation

- “Clearing the market” = accept/dispatch certain bids and offers, reject others
- Done by a central authority
- Objective: Maximize benefit to both generators and loads
 - First requirement: Authority is all-knowing of network, participants, bids/offers
 - Second requirement: One, unified decision
- Central authority is usually a government non-profit
 - Government authority + separate from generation/utility companies = fair market operation
 - Independent System Operators (ISOs), for example:
 - California ISO (CAISO)
 - Independent Electricity System Operator (IESO)

Different Types of Markets

- Wholesale electricity market
 - Day-to-day operation
 - Real-time operation
- Capacity market
 - Long-term capacity planning

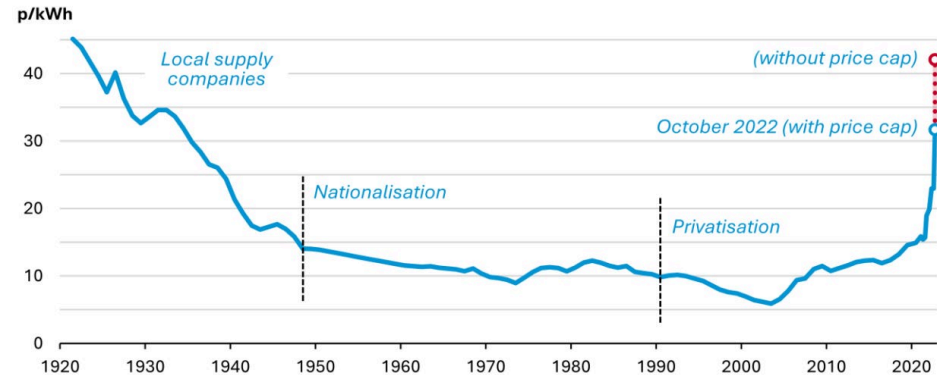


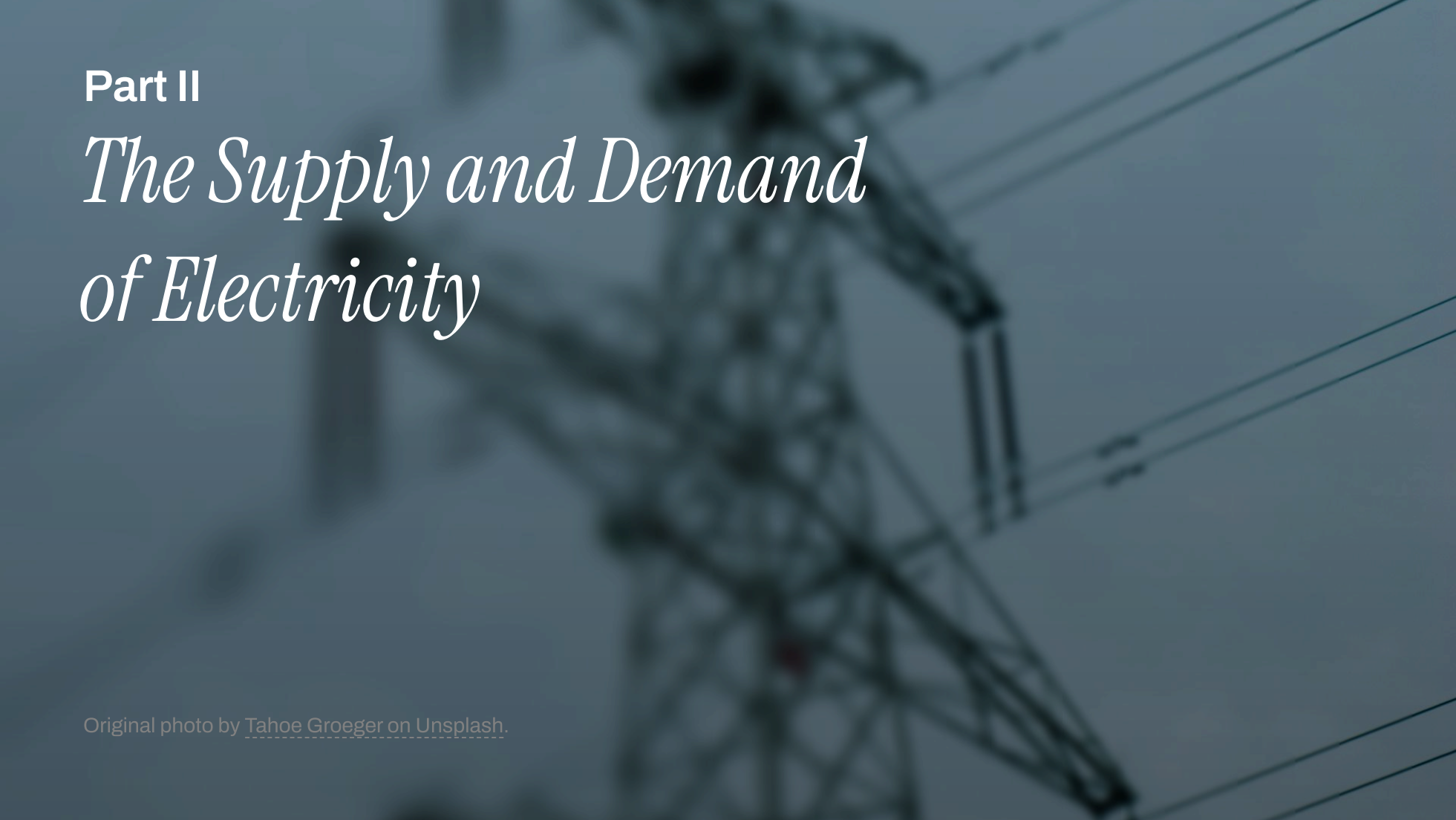
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Open Vs. Closed Markets

- System operators should be separate from generation/utility companies
- Generation/utility companies can still be private or government-owned
- Energy liberalization at turn of millennium:
 - Governments opened markets to private generation companies
 - Goal: competition $\Rightarrow$ lower prices
 - AKA “privatization”, “deregulation”
 - Strict market rules still in place
 - Government restructuring, e.g., IESO founded 1998

The retail price of electricity in Britain in real terms (2020 p/kWh), averaged across all consumers (residential, commercial and industrial).





Part II

The Supply and Demand of Electricity

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Introduction to Economics

Your Friendly Neighborhood Espresso Market

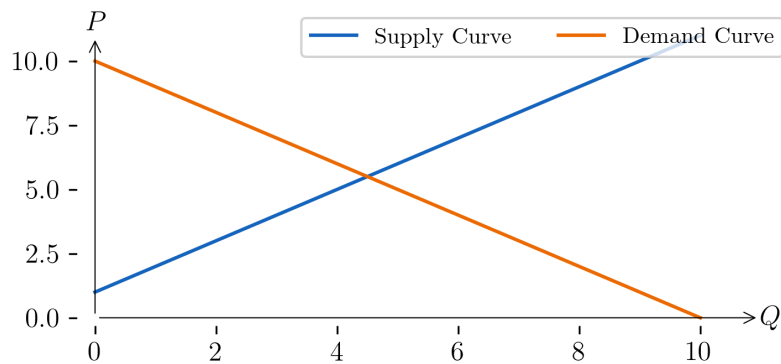
- Analogy: neighborhood of cafés
- Rate of espresso production = rate of espresso consumption
- Assume all cafés serve identical espresso, only differentiator is price:
 - Café undercuts $\Rightarrow$ others follow
 - Café charges more $\Rightarrow$ others follow
- At a given time and place: one stable espresso price

Photo by [Blake Wisz](#) on [Unsplash](#).



Supply and Demand Dynamics

- Demand:
 - Rate of consumption = $f(\text{price per shot})$
 - Higher price $\Rightarrow$ fewer customers
 - **Demand curve** (buyer behavior)
- Supply:
 - More espresso $\Rightarrow$ proportionately more spent on espresso beans, labor
 - A lot more espresso $\Rightarrow$ disproportionately higher:
 - Profit (price gauging)
 - Labor costs and production costs in general
 - Price per shot = $f(\text{rate of production})$
 - **Supply curve** (seller behavior)



Clearing Markets

- Quantity Q of a good is exchanged at price P
- At equilibrium: market-clearing quantity Q^* is exchanged at market-clearing price P^*
- The market rewards:
 - Sellers who charge a low price
 - Buyers who are willing to pay a high price
- Market clearing process:
 - Allocate buyers to sellers, starting with highest-price buyers and lowest-price sellers
 - Determines buying price = selling price = P^*

Graphic by [Nick Brunner on Unsplash](#).

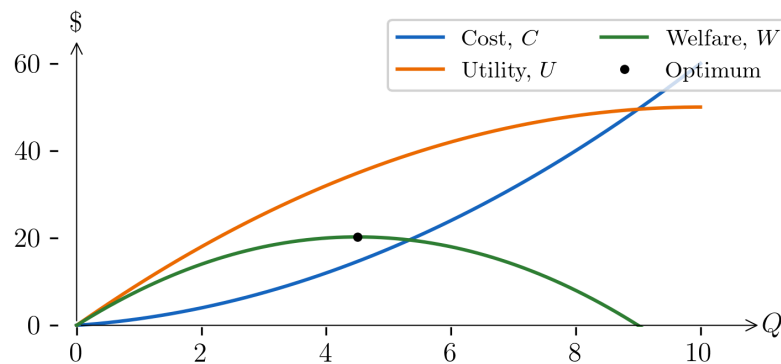
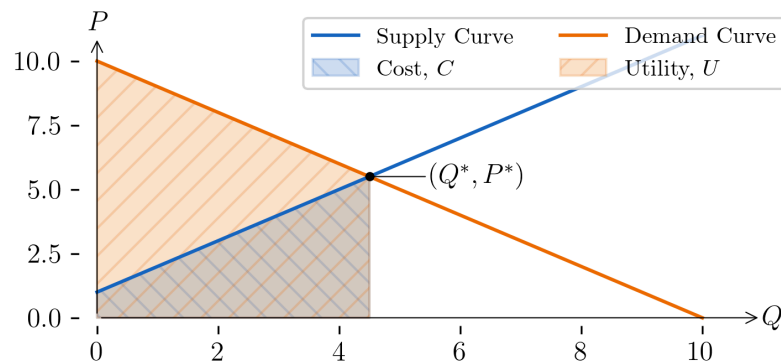


Clearing Markets

- P^*, Q^* determined by where **supply** and **demand** curves intersect
 - Area under **supply curve** = cost, C
 - Area under **demand curve** = utility, U
 - Constraint: $Q \text{ sold} = Q \text{ bought} = Q^*$
 - Maximizes U and minimizes C
- or

Maximizes **welfare**, $W = U - C$

- $W = \text{producer surplus} + \text{consumer surplus}$



Café-Grid Analogy

	In the café analogy	In the electrical grid
Good:	Espresso shots	Megawatt-hours (MWh)
Quantity unit:	Espresso shots per hour	Megawatts (MW)
Producers:	Espresso machines	Generators
Consumers:	Café customers	Loads

Electricity as a Commodity

- Commodity goods:
 - Espresso? Not a chance.
 - Wheat
 - Crude oil
 - **Electricity** – regardless of energy source
 - Nuclear
 - Hydro
 - Etc.

The Electricity Market Cannot Clear Itself

	Cafés	Electricity market
Buffer between production and consumption?	Yes – customers can wait for their orders to be fulfilled	No – loads simply take, without having to wait for power
Resources pooled?	No – each order is fulfilled by one barista	Yes – Increases in load can be shared across multiple generators



Photo by [Katt Galvan on Unsplash](#).

Role of the System Operator

- Monitors demand
- Dispatches generators to balance supply and demand
- Nice-to-have: independence from generators (independent system operator)
 - Creates competition
 - Reduces the cost of electricity

Time Variance

- Seen in supply/demand curves, quantities, and prices
- Fluctuations and spikes in supply/demand
→ fluctuations and spikes in electricity prices
- Price changes are characteristic of electricity markets



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Inelastic Loads

	Café customer	Non-dispatchable load
Elastic consumer?	Yes – aware of and responsive to café prices	No – utility customers are unresponsive to real-time electricity prices

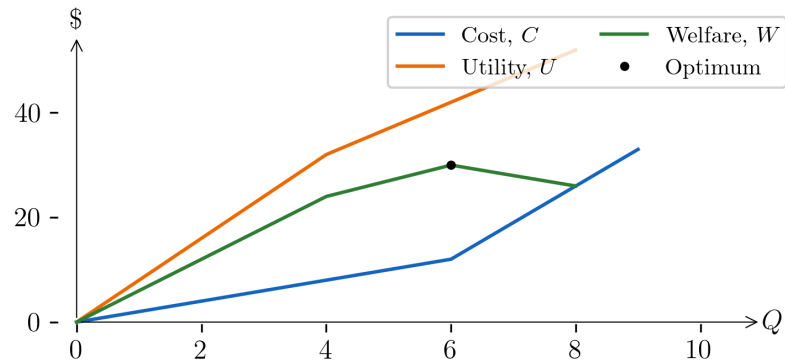
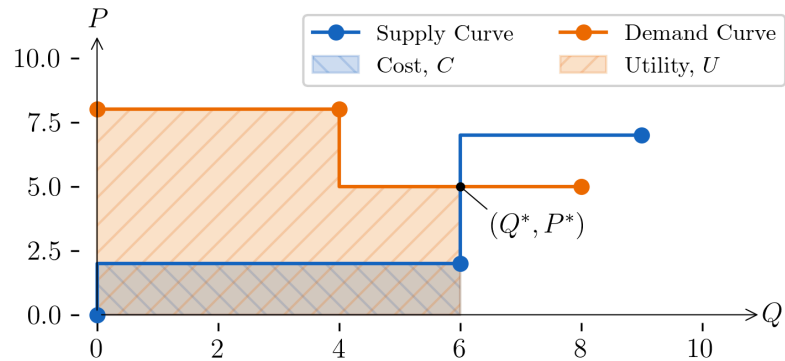
- Market participants:
 1. Non-dispatchable loads
 - Cause price spikes when demand is high or generation is intermittent
 2. Dispatchable loads
 - Large industrial consumers
 - Utility companies with demand response programs
 3. Non-dispatchable generators
 4. Dispatchable generators

Bids to Buy and Offers to Sell Power

- Bid/offer = price-quantity pairs
 - Specifies max quantity (MW) at each price (\$/MWh)
 - Offer $P(Q)$ monotonically increasing
 - Bid: $P(Q)$ monotonically decreasing

Example 1: One generator, one load ►

- Load's price-quantity pairs = demand curve
- Generator's price-quantity pairs = supply curve
- Intersection determines (Q^*, P^*)
- Maximizes $W = U - C$



Example 2: Two Generators, Two Loads

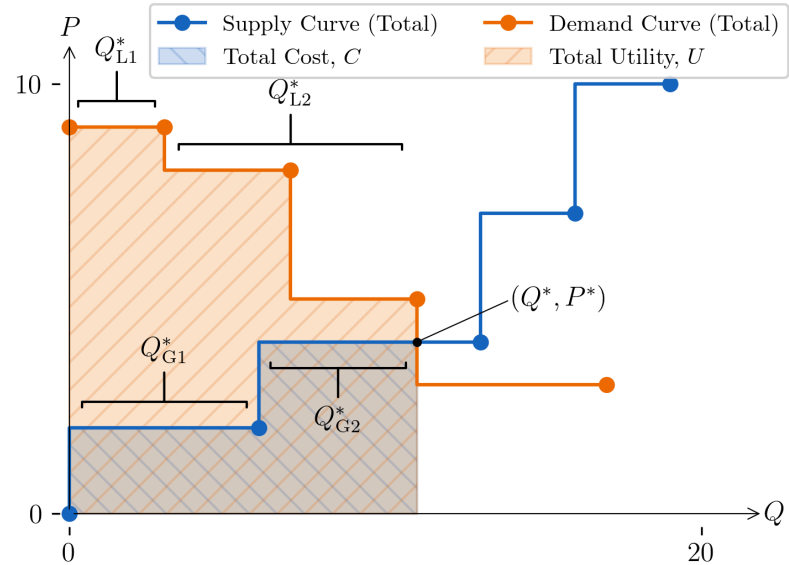
- Individual **supply** / **demand** curves
- Goal: maximize welfare, $W = U - C$
- Total seller cost:

$$C(Q_{G1}, Q_{G2}) = \int_0^{Q_{G1}} p_{G1}(q) dq + \int_0^{Q_{G2}} p_{G2}(q) dq$$

- Total buyer utility:

$$U(Q_{L1}, Q_{L2}) = \int_0^{Q_{L1}} p_{L1}(q) dq + \int_0^{Q_{L2}} p_{L2}(q) dq$$

- Economic Dispatch (ED) problem:**
 - $\max_{Q_{G1}, Q_{G2}, Q_{L1}, Q_{L2}} W(Q_{G1}, Q_{G2}, Q_{L1}, Q_{L2})$
 - subject to $Q_{G1} + Q_{G2} = Q_{L1} + Q_{L2}$
- Solution: $Q^* = 11$ MW, $P^* = \$4/\text{MWh}$,
 $C^* = \$32.0/\text{h}$, $U^* = \$79.0/\text{h}$, $W^* = \$47.0/\text{h}$





Part III

*Locational
Marginal Pricing*

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The Market Network

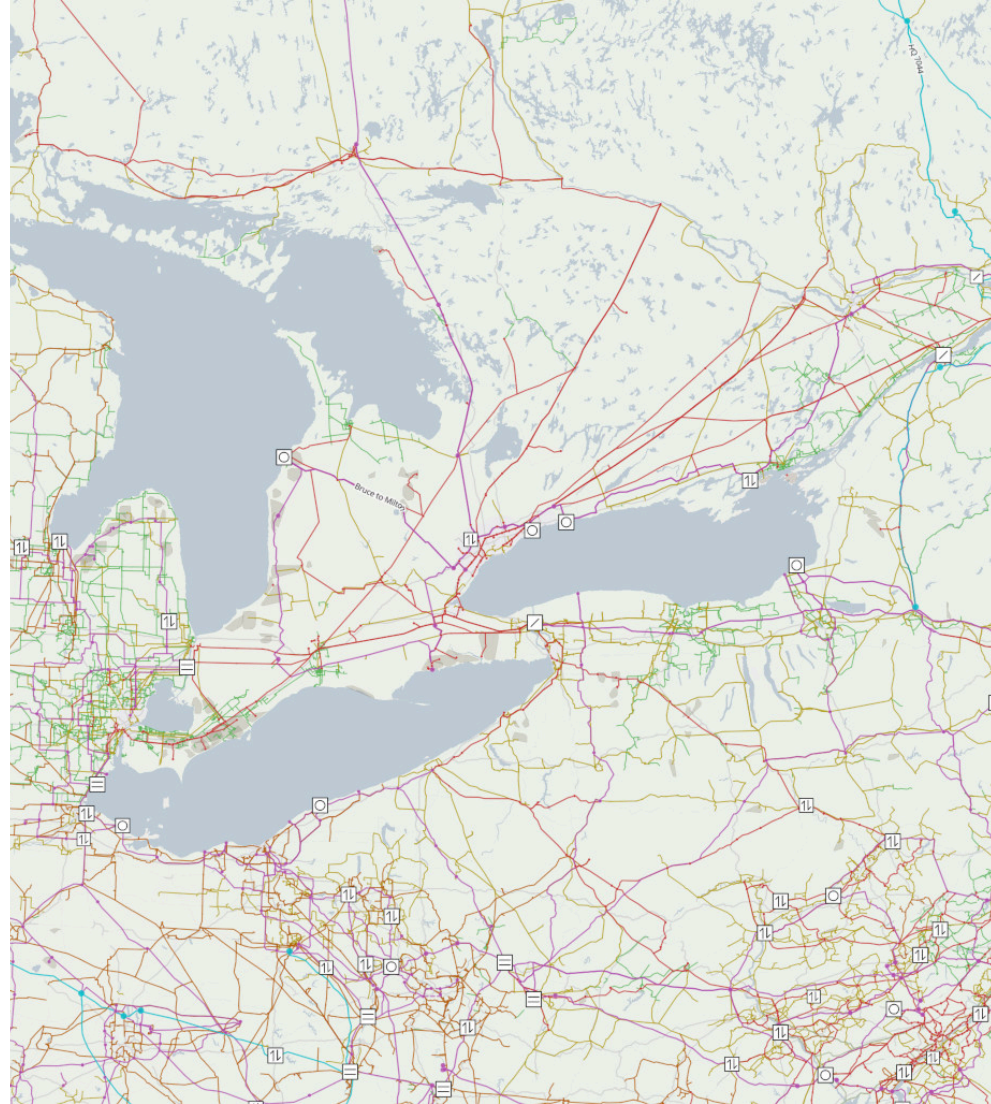
- One market price for a given time and market
- **Nodal pricing / locational marginal pricing (LMP):**
 - Grid = multiple pricing nodes, each like a market with its own price
- **Zonal pricing:**
 - Grid level → zonal level → nodal level
 - Pricing is at the zonal level

Grid infrastructure spanning Michigan, Ontario, and New York state ►

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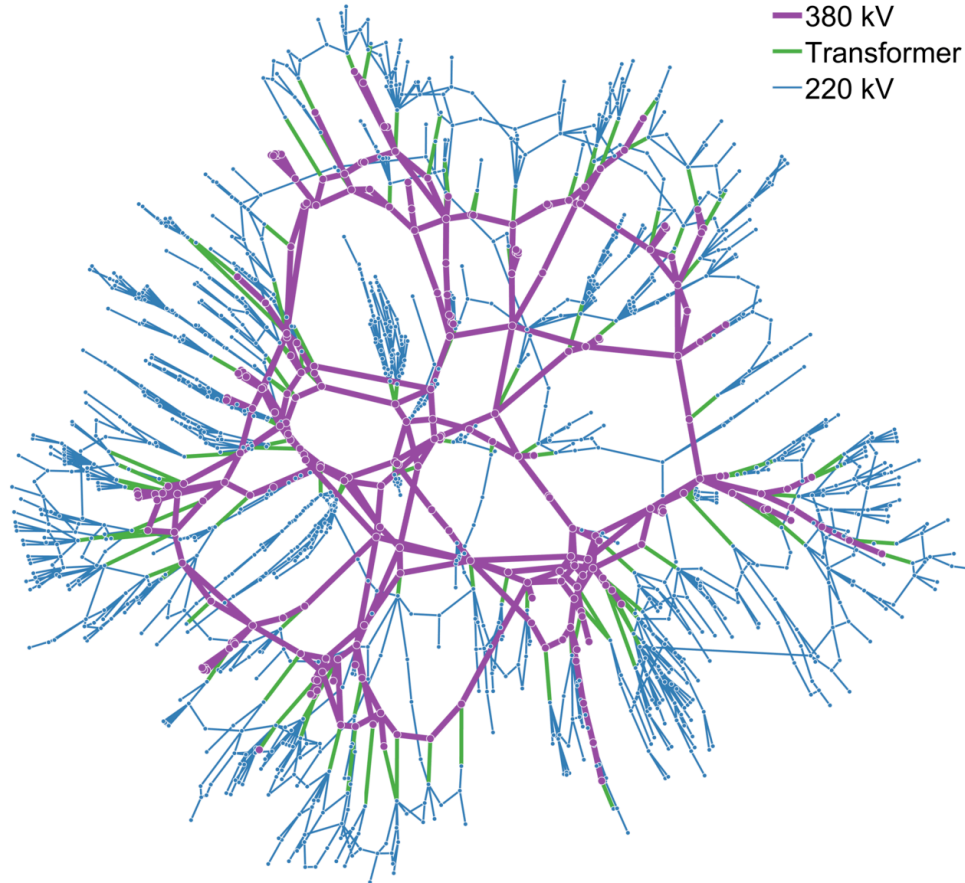
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Power Transmission

- Inter-node power transfer via transmission and distribution lines
- Supply must equal demand at the system level, not necessarily at the nodal level
- Analogous to trade
- The entire grid must be cleared as one market
- Maximizes total welfare
 - Higher with trade than without $\Leftrightarrow$ People trade to increase their welfare



Transmission Losses

- Transmission and distribution lines have electrical resistance
- $\text{Power in} = \text{power out} + \text{power lost to heat}$
- $\text{Power lost to heat} = \text{cost of trade}$
- Power transfer usually still beneficial to overall welfare
- Transmission losses cause price differences between nodes
- Simple model for the sake of our examples: efficiency $\eta = \text{power out} / \text{power in}$
- In the real world: more complex transmission line models

Transmission Congestion

- How a line outage occurs:
 - Thermal expansion
 - Line sags too close to nearby object (e.g., a tree)
 - Line shorts
 - Protective relay detects overcurrent and trips
- How line outages are avoided:
 - Rated maximum power
- Line is congested if $\text{power} = \text{maximum power}$
- Transmission congestion causes price differences between nodes

Example 3: One Generator Node, One Load Node

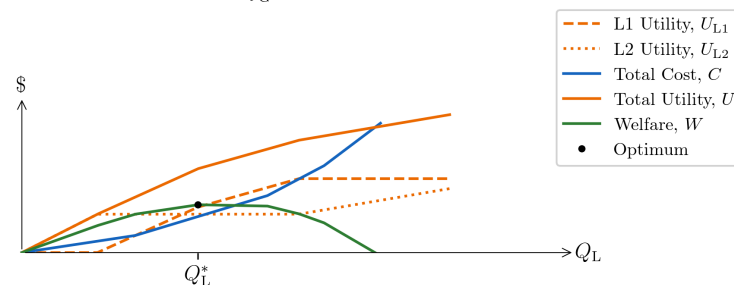
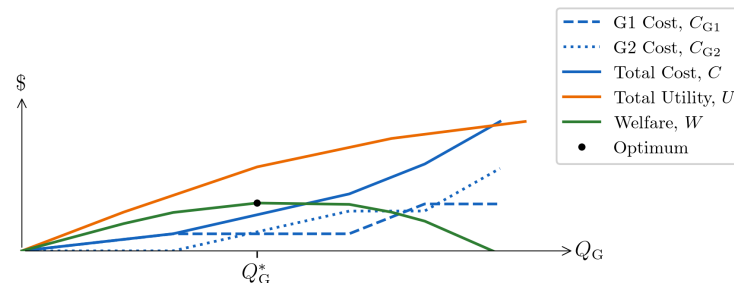
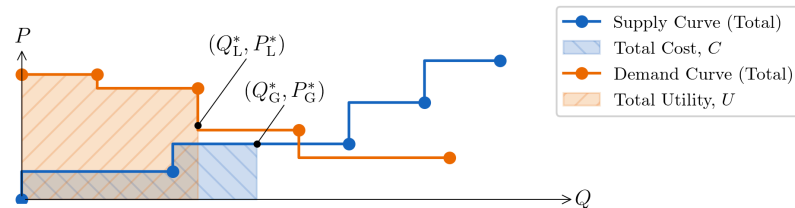
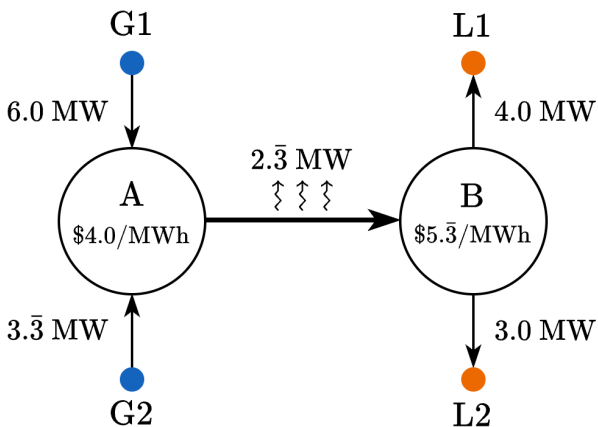
- Problem: $\max_{Q_G, Q_L} W(Q_G, Q_L)$

subject to:

- $Q_L = 0.75 Q_G$

- Solution:

$$C^* = \$25.3/\text{h}, U^* = \$59.0/\text{h}, W^* = \$33.6/\text{h}$$



Example 3 with Transmission Congestion

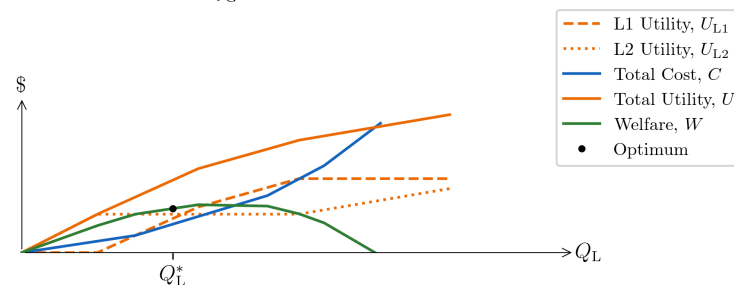
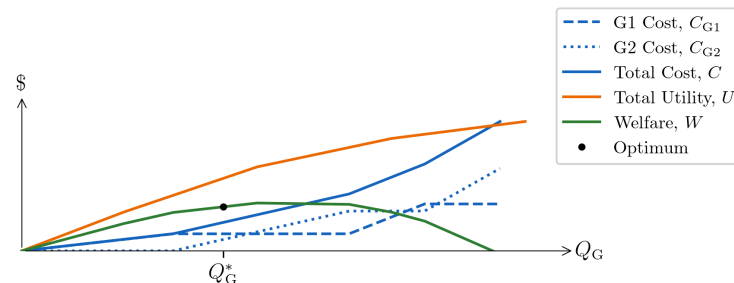
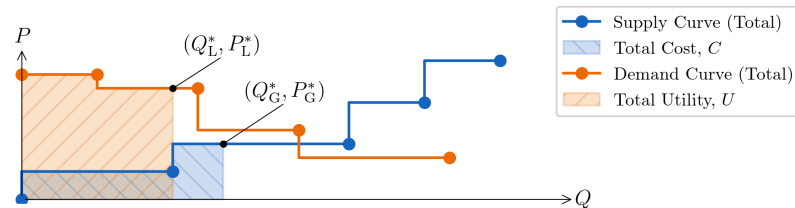
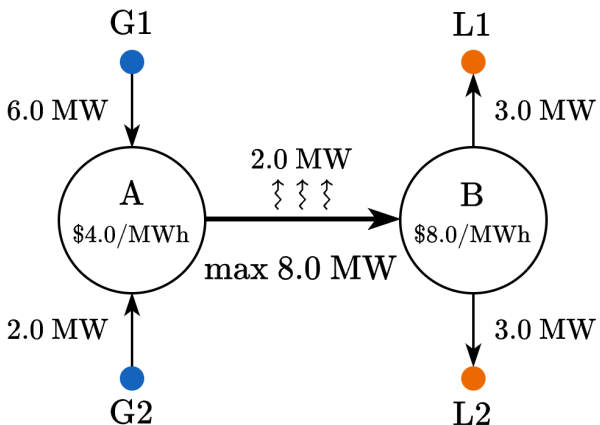
- Problem: $\max_{Q_G, Q_L} W(Q_G, Q_L)$

subject to:

- $Q_L = 0.75 Q_G$
- $Q_G < 8 \text{ MW}$

- Solution:

$$C^* = \$20.0/\text{h}, U^* = \$51.0/\text{h}, W^* = \$31.0/\text{h}$$



Example 4: One Generator Node, One Combined Generator/Load Node

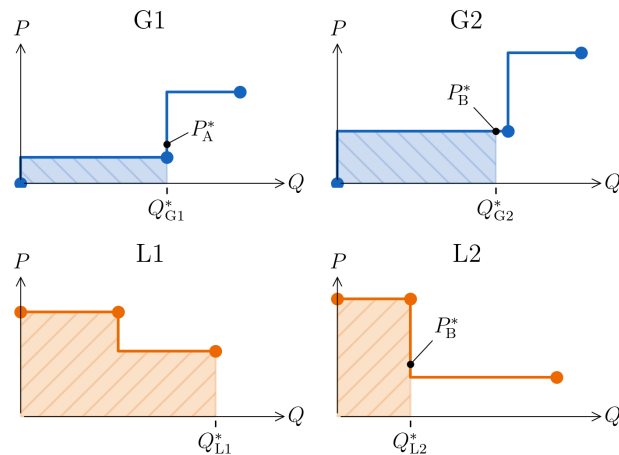
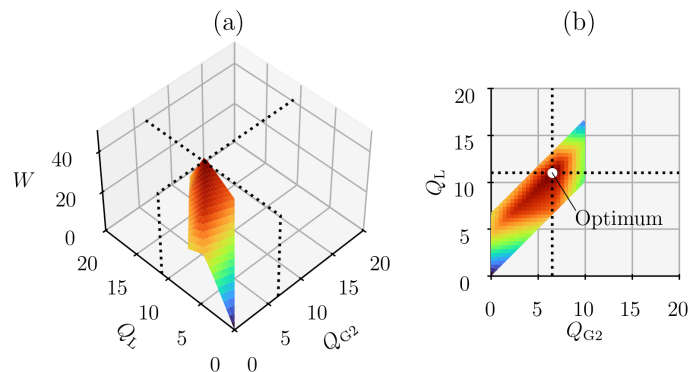
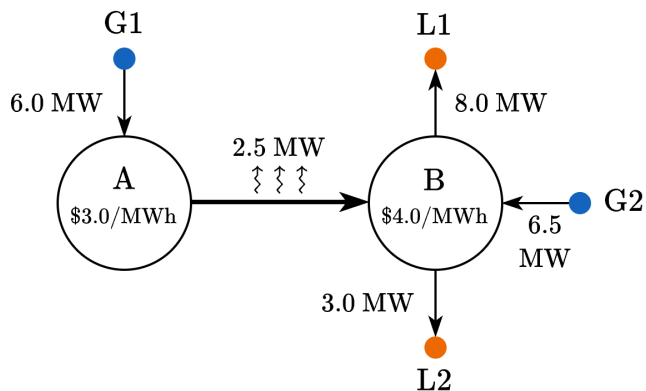
- Problem: $\max_{Q_{G2}, Q_L} W(Q_{G2}, Q_L)$

subject to:

- $0.75 Q_{G1} + Q_{G2} = Q_L$

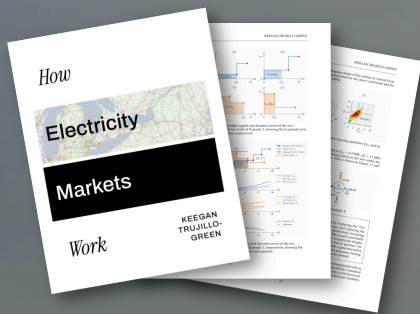
- Solution:

$$C^* = \$38.0/\text{h}, U^* = \$79.0/\text{h}, W^* = \$41.0/\text{h}$$



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